METRIC
DOD-S-70338
22 July 1988

MILITARY SPECIFICATION

SEAL RING, METAL, METRIC

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for metrically dimensioned seal rings.

1.2 Part number. The part number shall be as indicated on the applicable specification sheet (See 6.2 and applicable specification sheet).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESC-S, Picatinny Arsenal, N.J. 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5330

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MILITARY

| | |
|---|---|
| DOD-S-70338/1 | - Seal Ring, Copper, Metric |
| DOD-S-70338/2 | - Seal Ring, Aluminum, Metric |
| DOD-S-70338/3 | - Seal Ring, Steel, Metric |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-130 | - Identification Marking of U.S. Military Property |

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM

| | |
|---|---|
| ASTM A 568M | General Requirements for Steel, Carbon and High-strength Low-Alloy Hot-Rolled sheet and Cold-Rolled sheet (Metric) |
| ASTM B 133M | Standard Specification for Copper Rod, Bar and Shapes (Metric) |
| ASTM B 209M | Aluminum and Aluminum - Alloy Sheet and Plate (Metric) |
| ASTM E 8 | Standard Methods of Tension Testing of Metallic Materials |



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(Applications for copies of ASTM publications should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Material. The sealing rings shall be fabricated from the following materials as specified.

3.2.1 Copper. Copper shall be in accordance with ASTM B133M and shall exhibit properties as specified in Table I.

TABLE I. Properties of copper.

| Thickness mm | Strength Properties | | | | | Conductivity at 20° C minimum |
|---|---|---|---|---|---|---|
| | Tensile Strength MPa | 0.2 Offset Limit MPa | Elongation 1/ (% minimum) | | 2/ Brinell Hardness | |
| | | | $\delta_5$ | $\delta_{10}$ | | |
| 0.1 to 1 | 196 to 245 | 118 | 38 | 32 | 45 to 70 | 57 |
| over 1 | | | 45 | 38 | | |

1/ $\delta_5$ indicates elongation measured using a 5cm gage length.
$\delta_{10}$ indicates elongation measured using a 10cm gage length.

2/ Brinell hardness shall be measured using a ball with a diameter of 2.5mm and a load of 613N.

3.2.2 Aluminum. Aluminum shall be in accordance with ASTM B209M and shall exhibit properties as specified in Table II.



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TABLE II. Properties of aluminum.

| Thickness | | Strength Properties | | | | | |
|---|---|---|---|---|---|---|---|
| Sheet mm | Strip mm | Tensile Strength MPa | 0.2 offset limit MPa | Elongation 1/ % minimum $\sigma_5$ | Elongation 1/ % minimum $\sigma_{10}$ | 2/ Brinell Hardness | Conductivity at 20° C minimum |
| to 20 | to 3 | 70 to 100 | maximum 50 | 33 | 28 | 18 to 30 | 35.4 |
| to 6 | to 3 | 100 to 140 | minimum 80 (to 125) | 7 | 5 | 25 to 35 | 34.8 |
| to 4 | to 2.5 | 130 to 170 | minimum 110 (to 160) | 3 | 2 | 32 to 48 | 34.5 |
| to 1.5 | to 1.5 | minimum 160 | minimum 140 | 2 | 1 | 37 to 52 | 34.5 |

1/ $\sigma_5$ indicates elongation measured using a 5cm gage length.
$\sigma_{10}$ indicates elongation measured using a 10cm gage length.

2/ Brinell hardness shall be measured using a ball with a diameter of 2.5mm and a load of 613N.

3.2.3 Steel. Steel shall be as in accordance with ASTM A568M and shall exhibit properties as specified in Table III.

TABLE III. Properties of steel.

| Tensile Ident. Code | Chemical Analysis in % (maximum) | | | | Tensile Strength MPa | Yield Point MPa | Elongation % minimum |
|---|---|---|---|---|---|---|---|
| | C | P | S | N | | | |
| A | 0.20 | 0.08 | 0.05 | ----- | 362 to 440 | 216 | 18 |
| B | 0.10 | ---- | ---- | 0.007 | 270 to 368 | 250 | 32 |
| C | 0.20 | 0.06 | 0.50 | 0.008 | 362 to 440 | 216 | 18 |
| D | 0.20 | 0.06 | 0.05 | ----- | 362 to 440 | 216 | 18 |
| E | 0.25 | 0.06 | 0.05 | ----- | 412 to 490 | 235 | 16 |
| F | 0.30 | 0.06 | 0.05 | ----- | 490 to 588 | 294 | 14 |
| G | 0.20 | 0.05 | 0.05 | ----- | 510 to 608 | 353 | 16 |
| H | 0.40 | 0.06 | 0.05 | ----- | 588 to 706 | 334 | 10 |
| I | 0.50 | 0.06 | 0.05 | ----- | 686 to 834 | 363 | 6 |



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3.3 Surface hardness. The seal rings shall be fabricated having Brinell Hardness (BH) as indicated in Tables I and II.

3.4 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet (see 3.1).

3.5 Identification marking. The seal rings shall be identified in accordance with MIL-STD-130.

3.6 Workmanship. Seal rings shall be free from surface contamination, tool marks and other imperfections which may adversely affect useability.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.1 and 4.6.2.



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4.4 Materials inspection. Materials shall be tested in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents. The sample selected shall be suitably marked and dated for identification with the inspection lot. Samples for testing may be selected from the stock material from which the end item is made at the time of manufacture. The stock material shall be unacceptable if one or more sample units fail to meet any test requirements specified (see 4.1).

4.4.1 Material certification. The contractor shall provide to the government when requested certification verifying the use of proper materials defined in 3.2.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in Table IV.

TABLE IV. Quality conformance inspection.

| Inspection | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Group A Examination | 1/ | 4.6.1 |
| Group B Surface Hardness | 3.3 | 4.6.2 |

1/ All requirements not covered by tests.

4.5.1 Inspection lot. An inspection lot shall consist of all seal rings covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.5.3 Sampling for inspection. Sampling shall be in accordance with MIL-STD-105 Inspection Level III with an Acceptable Quality Level (AQL) of 0.25 percent defective for major defects. For minor defects Inspection Level I with an AQL of 1.0 percent defective shall apply.



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4.5.4 Classification of defects. Defects shall be classified as specified in Table V.

TABLE V. Classification of defects.

| Category | Defect |
|---|---|
| Major<br>101<br>102 | <br>Diameter (see 3.4)<br>Thickness (see 3.4) |
| Minor<br>201<br>202<br>203<br>204 | <br>Incorrect marking identification (see 3.5)<br>Workmanship not as specified (see 3.6)<br>Packaging not as specified (see 5.1)<br>Other dimensional characteristics not as covered above |

4.5.5 Inspection of packaging The sampling and inspection of the preservation -packaging, packing and container marking shall be in accordance with PPP-H-1581.

4.6 Methods of inspection.

4.6.1 Examination. The seal rings shall be examined for defects in design, appearance and workmanship as specified in Table IV. Sample seal rings shall be thoroughly examined to determine conformance with this specification and the applicable specification sheet with respect to all the requirements not covered by tests.

4.6.2 Surface hardness. Sample seal rings shall be tested for surface hardness (see 3.3).

4.6.3 Mechanical properties. The tensile strength elongation and yield point shall be as specified in Tables I, II and III and shall be tested accordance with ASTM E8.



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## 5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

## 6 NOTES

6.1 Intended use. Seal rings acquired under this specification are intended for use in pipe unions and screw plugs.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.

b. Applicable sizes in accordance with specification sheet part number(s) (see 3.1).

c. Applicable levels of preservation, packing and marking (see 5.1).

6.3 Subject term (key word) listing.

a. seal
b. ring
c. metric



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Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Army - MI, MR
Navy - OS
Air Force - 82
DLA - IS

User activities:
Army - AT, AV, GL, ME
Navy - AS, YD
Air Force - 68

Civil agencies and coordinating activities:
GSA - FSS

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5330-0685)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| DOD-S-70338 | SEAL RING , METAL, METRIC |

| 3. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6 REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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*NOTE*: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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